

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V) Data
Visualization with Power BI and Tableau

Practical – 9

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Aim : Cleaning and Structuring Data for Analysis

A) Identify and handle missing or inconsistent data:

Step 1: Load the Dataset

- . Open Tableau Desktop and load the given dataset.
- . Go to Sheet 1.

Step 2: Clean Physics and Attendance

- . Right-click in the Data Pane - Create Calculated Field.

. Create Physics_Clean.

`FLOAT (REPLACE (STR( [Physics] ), '%', ''))`

. Repeat the same steps and create Attendance_Clean.

`FLOAT (TRIM (REPLACE (STR ( [Attendance] ), '%', '')))`

Physics_clean

```

FLOAT (REPLACE (STR ( [Physics] ), '%', ''))

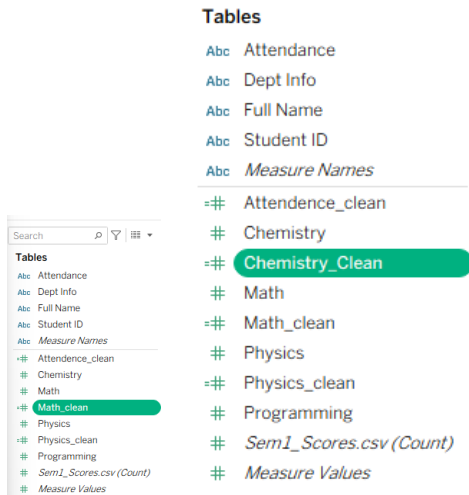
```

The screenshot shows the Tableau Desktop interface. On the left, the 'Create Calculated Field' dialog box is open, displaying various options for creating new fields. On the right, the 'Data Pane' lists several data sources, including 'Attendance', 'Dept Info', 'Full Name', 'Student ID', 'Measure Names', 'Chemistry', 'Math', 'Physics', 'Physics_clean', 'Programming', 'Sem1_Scores.csv (Count)', and 'Measure Values'. The 'Physics_clean' table is highlighted in green, indicating it is the selected data source for the current calculation.

Step 3: Handle Missing Values

. Create Math_Clean using `IFNULL([Math], 0)`.

. Create Chemistry_Clean using `IFNULL([Chemistry], 0)`.



Math_clean

```
IFNULL([Math], 0)
```

Chemistry_Clean

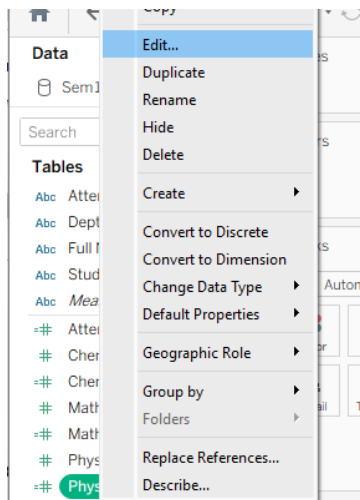
```
IFNULL([Chemistry], 0)
```

Math	▼	Create Calculated Field...
Physics	▼	
Chemistry	▼	Pivot
Programming	▼	Merge Mismatched Fields

Step 4: Handle Missing Physics Values

. Right-click Physics_Clean - Edit and use:

Formula: `IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)`



Field Name	Physical Tab
Dept Info	Sem1_Score
Attendance	Sem1_Score
Pivot Field Names	Pivot
Pivot Field Values	Pivot
Physics_clean !	Calculation
Attendance_clean	Calculation
Math_clean !	Calculation

Step 3: Rename Pivoted Fields

. name Pivot Field Names to Subject.

. name Pivot Field Values to Score.

Physics_clean

```
IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)
```

B) Convert Wide Data into Tall Format:

Step 1: Open Data Source . Open the Data Source tab.

Step 2: Pivot Columns:

. Hold Ctrl and select Math, Physics, Chemistry, Programming.

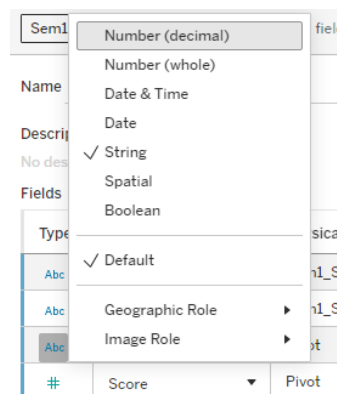
. Right-click the selected columns - Pivot.

Abc	Attendance	▼	S
Abc	Subject	▼	P
#	Score	▼	P
#	Physics_clean !	▼	C
#	Attendance_clean	▼	C

Step 4: Format Score

Click the Abc icon above Score.

Select Number (decimal).

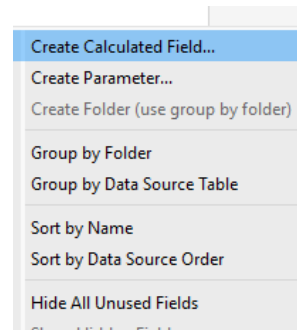


Alternative: Create Clean_Score

Formula: IFNULL (FLOAT (REPLACE (STR([Score]), '%', '')), 0)

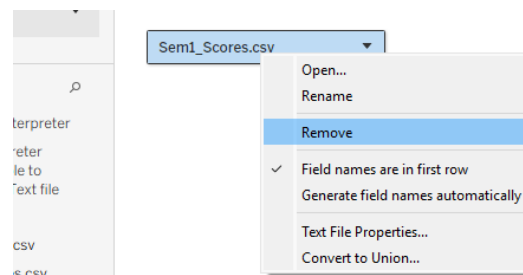
Clean_Score

```
IFNULL (FLOAT (REPLACE (STR([Score]), '%', '')), 0)
```



C) Perform Union of Multiple Files:

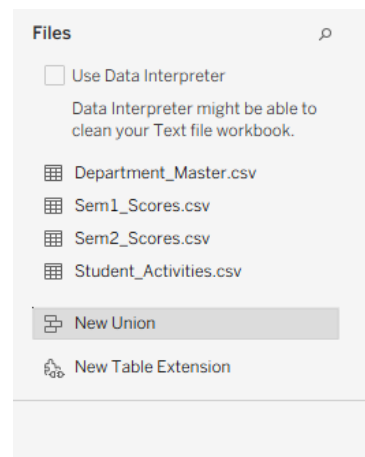
Step 1: Open the Data: Clear the existing canvas



Step 2: Create a Multi-File Union

. Drag New Union to the canvas.

. Add Sem1_Scores.csv and Sem2_Scores.csv.



Connection: Sem1_Scores



Tables in union: 2



3: Verify the Merged Data Grid

Check the data grid at the bottom.

Confirm that the total is 20 rows.

Check the Table Name column for the two files.



Scroll to the far right of the grid to verify the generated system dimension: Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and

"Sem2_Scores.csv" for the remaining 10 rows.

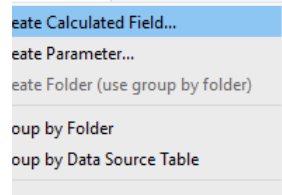
	Abc Union	Calculation	Calculation
Table Name	Physics_clean	Attendance	
Sem1_Scores.csv	0.900000		
Sem1_Scores.csv	0.750000		
Sem1_Scores.csv	0.820000		
Sem1_Scores.csv	0.880000		
Sem1_Scores.csv	0.000000		
Sem1_Scores.csv	0.910000		
Sem1_Scores.csv	0.700000		

4: Create a Clean Semester Identifier Dimension

Click on Sheet 1 at the bottom-left.

Create Calculated Field.Set the Field Name to: Semester

Enter the following formula:



```
CONTAINS([Table Name], "Sem1") THEN "Semester 1"
SEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
SE "Other"
```

Execute cross-database joins:

1: Stay on the Top Canvas Leave your Union table box as it is in the canvas.

2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv. 2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.



4. If the relationship dialog opens (or click the relationship line):
 - o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =

Union — Department...

Name Department_Master.csv

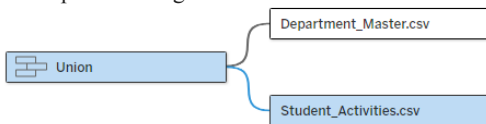
How do relationships differ from joins? [Learn more](#)

Union	Operator	Department_Maste...
Dept_Info	=	Dept_Code



Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.



3. In the relationship dialog:
 - o Left Table Field (Union): Select Student_ID
 - o Right Table Field (Student_Activities): Select Student_ID
 - o Operator: =

Union — Student_A...

Name Student_Activities.csv

How do relationships differ from joins? [Learn more](#)

Union	Operator	Student_Activities....
Student_ID	=	Student ID (Stu...

E. Restructure Data for Better Visualization

Step 1: Calculate Total Marks

1. Open Tableau.
2. Go to Sheet 1.
3. In the Data Pane, right-click on an empty area.
4. Select Create Calculated Field.
5. Enter field name: Total_Score
6. Enter formula: [Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]
7. Click OK.

Step 4: Verify in Sheet 1

Click on Sheet 1 at the bottom-left.

In the Data Pane on the far left, you will now see all three tables organized with their respective fields:

- o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
- o Department_Master: Dept_Code, Department_Name, Dept_Name, Dept_Target_Average, Total_Dept_Capacity
- o Student_Activities: Club_Membership, Projects_Completed, Certifications

bles	
Department_Master.c...	
Abc Department Name	
Abc Dept Code	
Abc HOD Name	
# Dept Target Average	
# Total Dept Capacity	
# Department_Master...	
Student_Activities.csv	
Abc Club Membership	
Abc Student ID (Student...	
# Certifications	
# Projects Completed	
# Student_Activities.c...	
Union	
Abc Attendance	
Abc Dept_Info	
Abc Full_Name	
Abc Semester	
Abc Student_ID	
Abc Table Name	
Attendance_clean	
Chemistry	
Chemistry_Clean	
Math	
Math_clean	
Physics	
Physics_clean	

total_score

[Math] + [Physics] + [Chemistry] + [Programming]

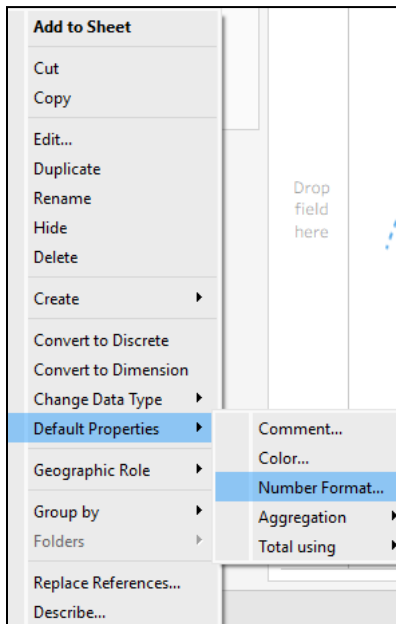
Step 2: Calculate Overall Percentage

1. Right-click in the Data Pane.
2. Select Create Calculated Field.
3. Field name: Overall_Percentage
4. Formula: [Total_Score] / 400.0
5. Click OK.

- Right-click Overall_Percentage.
- Select Default Properties → Number Format.
- Select Percentage.
- Set decimal places to 1.
- Click OK.

```
total_score
```

```
[Math] + [Physics] + [Chemistry] + [Programming]
```



Step 3: Create Performance Band

- Right-click in Data Pane.
- Select Create Calculated Field.
- Field name: Performance_Band

Enter:

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
```

- END
- Click OK.

```
Performance_Band1
```

```
IF [overall_percentage] >= 0.85 THEN "Distinction"
ELSEIF [overall_percentage] >= 0.75 THEN "First Class"
ELSEIF [overall_percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END
```

Step 4: Create Attendance Status

- Right-click Data Pane.
- Select Create Calculated Field.
- Field name: Attendance_Status

Formula: IF [Attendance_clean] >= 75 THEN
"Eligible (Good Standing)"
ELSE

"Shortage Warning (<75%)"

END

- END
- Click OK.

```
Attendance_Status
```

```
IF [Attendance_clean] >= 75 THEN
  "Eligible (Good Standing)"
ELSE
  "Shortage Warning (<75%)"
END
```

Step 5: Build Student Performance Scorecard

Now create the visualization:

- Drag Student_ID → Rows.
- Drag Full_Name → Rows.
- Drag Semester → Columns.
- Drag Overall_Percentage → Text in Marks card.
- Drag Performance_Band → Color.
- Drag Attendance_Status → Tooltip or Detail.

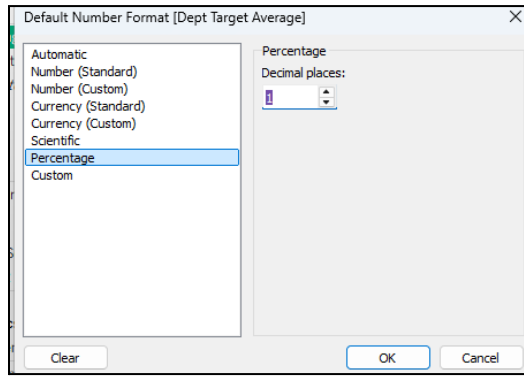
Output:-

Student ID	Full Name	Semester	Performance_Band	Attendance_Status	SUM(overall_p)
S101	Aarav Sharma	Semester 2	Distinction	Eligible (Good Standing)	67.2%
S102	priya VERMA	Semester 2	First Class	Eligible (Good Standing)	69.7%
S103	Rohan Gupta	Semester 2	Second Class	Eligible (Good Standing)	66.5%
S104	Sneha Patel	Semester 2	Needs Academic Support	Shortage Warning (<75%)	71.5%
S105	aditya Rao	Semester 2	Distinction	Eligible (Good Standing)	49.4%
S106	Ananya Singh	Semester 2	First Class	Eligible (Good Standing)	63.0%
S107	Rahul Nair	Semester 2	Second Class	Eligible (Good Standing)	
S108	Kavya Reddy	Semester 2	Needs Academic Support	Shortage Warning (<75%)	
S109	Vikram Joshi	Semester 2	Distinction	Eligible (Good Standing)	
S110	Pooja Iyer	Semester 2	First Class	Eligible (Good Standing)	

F. Handle Different Levels of Detail Using LOD

Step 1: Department Average

- Go to Sheet 1 or create Sheet 2.
- Right-click in Data Pane.
- Select Create Calculated Field.
- Field name: Dept_Average_Percentage_LOD
- Formula: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
- Click OK.
- Right-click the new field.
- Select Default Properties → Number Format.
- Select Percentage and 1 decimal place.



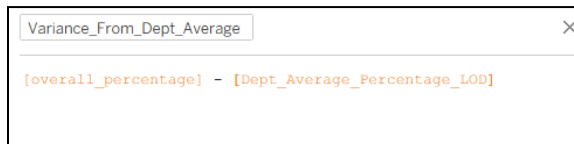
Step 2: Student Career Average

1. Right-click Data Pane.
2. Select Create Calculated Field.
3. Field name:
Student_Overall_Career_Average
4. Formula:
{ FIXED [Student_ID] :
AVG([Overall_Percentage]) }
5. Click OK.
6. Set the number format to Percentage with 1 decimal place.



Step 3: Calculate Variance

1. Create another Calculated Field.
2. Field name:
Variance_From_Dept_Average
3. Formula:
[Overall_Percentage] -
[Dept_Average_Percentage_LOD]
4. Click OK.
5. Set it to Percentage with 1 decimal place.



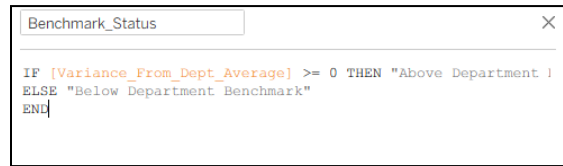
Step4: Create Benchmark Status

1. Create another Calculated Field.
2. Field name: Benchmark_Status

Formula:

```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
```

3. END
4. Click OK.



Step 5: Build LOD Visualization

1. Drag Dept_Info → Rows.
2. Drag Student_ID → Rows.
3. Drag Full_Name → Rows.
4. Drag Semester → Columns.
5. Drag Overall_Percentage → Text.
6. Drag Dept_Average_Percentage_LOD → Tooltip.
7. Drag Benchmark_Status → Color.

Final result

Pages

Filters

Marks

Columns

Rows

Semester

Dept Info

Student ID

Full Name

Automatic

Color

Size

Text

Detail

Tooltip

Benchmark_Status

SUM(Dept_Tar_

SUM(overall_p

Semester

Semeste

r 2

Dept Info

Student ID

Full Name

CSE-01

S101

Aarav Sharma

67.2%

S103

Rohan Gupta

S106

Ananya Singh

66.5%

S109

Vikram Joshi

49.4%

ECE-02

S102

priya VERMA

S107

Rahul Nair

S110

Pooja Iyer

63.0%

IT-03

S104

Sneha Patel

69.7%

S108

Kavya Reddy

71.5%

MECH-04

S105

aditya Rao

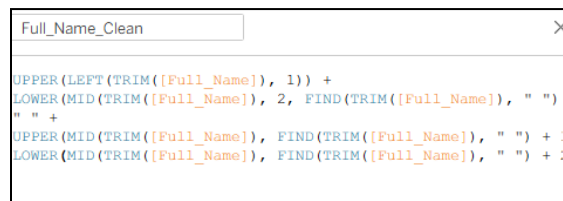
G: Apply Advanced Data Cleaning Techniques

STEP 1 — Full_Name_Clean

1. Sheet 1 par raho.
2. Left side Data Pane mein right-click karo.
3. Create Calculated Field select karo.
4. Field Name: Full_Name_Clean
5. Formula:

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +  
LOWER(MID(TRIM([Full_Name]), 2,  
FIND(TRIM([Full_Name]), " ") - 1)) +  
" " +  
UPPER(MID(TRIM([Full_Name]),  
FIND(TRIM([Full_Name]), " ") + 1, 1)) +  
LOWER(MID(TRIM([Full_Name]),  
FIND(TRIM([Full_Name]), " ") + 2,  
LEN(TRIM([Full_Name]))))
```

6. OK click karo.



STEP 2 — First_Name

1. Data Pane → Right Click
2. Create Calculated Field

3. Name:
First_Name

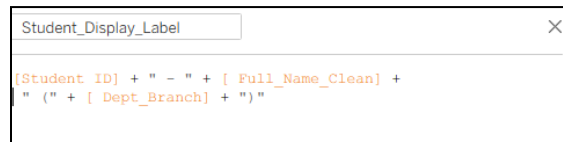
4. Formula:
TRIM(SPLIT([Full_Name], " ", 1))

5. OK.



A screenshot of a dialog box titled 'First_Name' with a close button (X) in the top right corner. The text area contains the formula: `TRIM(SPLIT([Full_Name], " ", 1))`.

4. OK.



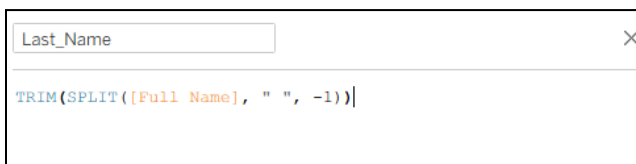
A screenshot of a dialog box titled 'Student_Display_Label' with a close button (X) in the top right corner. The text area contains the formula: `[Student ID] + " - " + [Full_Name_Clean] +
" (" + [Dept_Branch] + ")"`

STEP 3 — Last_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name:
Last_Name

4. Formula:
TRIM(SPLIT([Full_Name], " ", -1))

5. OK.



A screenshot of a dialog box titled 'Last_Name' with a close button (X) in the top right corner. The text area contains the formula: `TRIM(SPLIT([Full Name], " ", -1))`

STEP 4 — Dept_Branch

Tumhare Dept_Info mein values jaise CSE-01, ECE-02 hain.

1. Create Calculated Field
2. Name:
Dept_Branch

3. Formula:
SPLIT([Dept_Info], "-", 1)

4. OK.



A screenshot of a dialog box titled 'Dept_Branch' with a close button (X) in the top right corner. The text area contains the formula: `SPLIT([Dept Info], "-", 1)`

STEP 5 — Section_ID

1. Create Calculated Field
2. Name:
Section_ID

3. Formula:
SPLIT([Dept_Info], "-", 2)

4. OK.



A screenshot of a dialog box titled 'Section_ID' with a close button (X) in the top right corner. The text area contains the formula: `SPLIT([Dept Info], "-", 2)`

STEP 6 — Student_Display_Label

1. Create Calculated Field
2. Name:
Student_Display_Label

3. Formula:
[Student_ID] + " - " + [Full_Name_Clean] + " (" +
[Dept_Branch] + ")"